



DEPARTMENT OF THE AIR FORCE
HEADQUARTERS AIR FORCE SUSTAINMENT CENTER (AFMC)
TINKER AIR FORCE BASE OKLAHOMA

13 March 2026

TO: AFSC/PZIMB

FROM: OC-ALC/OBCA for 76 MXSG

SUBJECT: Brand Name Justification for M26-110 Handheld XRF Analyzer

1. This requirement is for a portable, handheld X-ray fluorescence (XRF) analyzer to conduct immediate, non-destructive elemental analysis of materials in field and laboratory environments. The acquired item, a Niton XL5 Plus handheld XRF analyzer, will be used to rapidly identify and verify the elemental composition of [REDACTED] assets, ensuring material integrity, safety, and [REDACTED].

2. The Niton XL5 Plus is the only instrument that meets the government's minimum requirements due to its unique and proprietary technology. The analyzer's powerful 5W X-ray tube, the most powerful available in a handheld device, is combined with a proprietary Geometrically Optimized Large Area Drift Detector (GOLDD+). This synergistic pairing delivers superior performance, enabling faster analysis times and more precise results compared to conventional analyzers.

A [REDACTED] and differentiating capability is the analyzer's ability to accurately quantify light elements—such as magnesium (Mg), aluminum (Al), silicon (Si), phosphorus (P), and sulfur (S)—without the need for cumbersome vacuum pumps or helium purge systems. Conventional analyzers from other manufacturers cannot perform this function under field conditions, as the weak fluorescent X-rays from light elements are typically absorbed by the air. This capability is [REDACTED] for identifying and verifying specific alloys and materials [REDACTED]. The GOLDD+ technology is a proprietary and trademarked innovation exclusive to the Thermo Fisher Scientific Niton line and is the only technology that can meet this requirement.

3. Therefore, it is requested that the Thermo Scientific Niton™ XL5 Plus brand name is procured to meet all the requirements of the program office [REDACTED]. This is the only known brand that can satisfy the Government's [REDACTED] requirement for rapid, field-based analysis of light elements without external equipment.

4. Market research was conducted to identify handheld XRF analyzers capable of meeting the government's requirements. This research confirmed that while several manufacturers produce handheld XRF analyzers, none offers the specific combination of a 5W X-ray tube and the proprietary GOLDD+ detector technology. This unique combination is the only solution capable of meeting the [REDACTED] need for analyzing light elements without external pumps or gas purge systems. The Government will continue to conduct market research for future acquisitions and will compete with this requirement if other vendors develop technology that meets the Government's minimum technical needs.

5. For further information or clarification please contact

[REDACTED]

[REDACTED]

6. I have determined, in accordance with FAR 13.106(b)(1), that the circumstances of this contract action deem a specific brand name to be required.

[REDACTED]